TRANSMISSION LINES & WAVEGUIDES – ECA1501

EXPERIMENT – 4

Strategize Impedance Matching With Characteristic Impedance And Load Impedance Through The Measurement Of A) Location Of The Stub B) Length Of The Stub

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Date	16/10/2027	Experiment No.	4

AIM

To design strategize impedance matching with characteristic impedance and load impedance through the measurement of a) location of the stub b) length of the stub.

SOFTWAREREQUIRED:

SCILABVersion6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clc;
clear;
Zo = 60;
f = 100 * 10^6;
lambda = 3 * 10^8 / f;
printf("ZR(Ohms)\tPoint of attachment(cm)\tStub length(cm)\n");
for ZR = 100:50:350
   theta = 30 * %pi / 180;
   ZL = ZR * (cos(theta) + %i * sin(theta));
```

```
    zL = ZL / Zo;
    yL = 1 / zL;
    g = real(yL);
    b = imag(yL);
    A = g - g^2 - b^2;
    B = -2 * b;
    C = g - 1;
    D = B^2 - 4 * A * C;
    if D >= 0 then
       t1 = (-B + sqrt(D)) / (2 * A);
       d = atan(t1);
       if d < 0 then
          d = d + %pi;
       end
       y = (yL + %i * tan(d)) / ...
          (1 + %i * yL * tan(d));
       bs = -imag(y);
       ls = atan(bs);
       if ls < 0 then
          ls = ls + %pi;
       end
       printf("%d\t\t%.2f\t\t\t%.2f\n", ...
       ZR, lambda*d/(2*%pi)*100, ...
       lambda*ls/(2*%pi)*100);
    else
       printf("%d\t\tNo real solution\n", ZR);
    end
end
```

TEST CASE 1

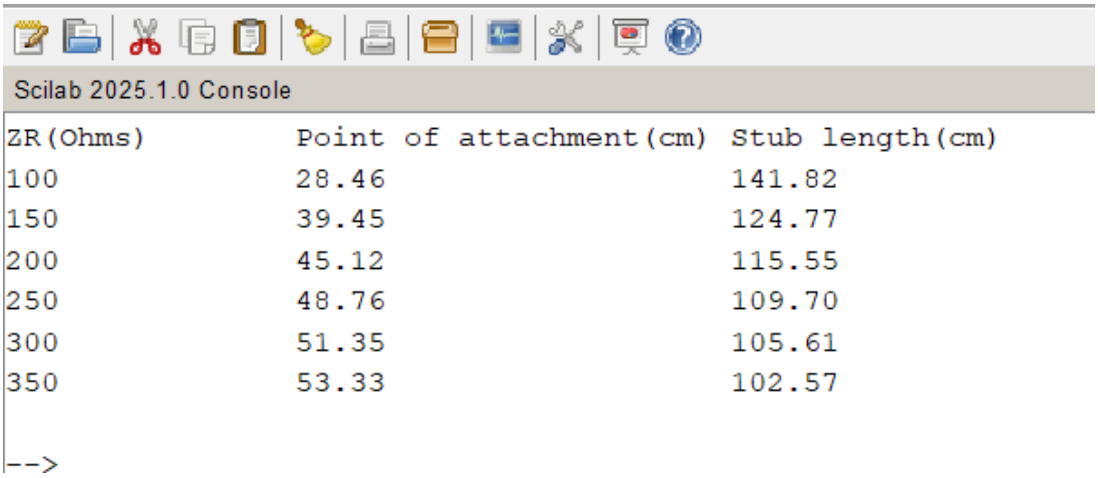
Optimize impedance matching using single stub matching by finding the location and length of the stub for the values of load impedance of 100 30o to 350 30o keeping the characteristic impedance Zo= 60 Ω constant .

OUTPUT TABLE 1

Variables ZR;
Constants Zo = 60Ω; f=100*(10^6);

Input	**Output**	
ZR Ohms. angle 30o	**Point of attachment**	**Length of short-circuited stub (cm)**
100∠30°	28.46	141.82
150∠30°	39.45	124.77
200∠30°	45.12	115.55
250∠30°	48.76	109.70

300∠30°	51.35	105.61



RESULT:

Impedance matching using single stub matching is optimized by calculating the location and length of the stub or varying values of load impedance keeping the characteristic impedance Zo= 60 Ω constant .

TEST CASE 2

Optimize impedance matching using single stub matching by finding the location and length of the stub for the values of characteristic impedance ranging from 600Ω to 1800 Ω. Keeping the load impedance ZR= 100 30o constant .

OUTPUT:

Variables Z0;
Constants ZR= 100 30o Ω; f=100*(10^6);

Input	**Output**	
Z0 Ohms. angle 30o	**Point of attachment**	**Length of short-circuited stub (cm)**
600	128.67	19.87
900	132.98	15.76
1200	135.48	13.44
1500	137.15	11.91
1800	138.37	10.81

Scilab 2025.1.0 Console

```
Z0(Ohms)          Point of attachment(cm)  Stub length(cm)
600               128.67                   19.87
900               132.98                   15.76
1200              135.48                   13.44
1500              137.15                   11.91
1800              138.37                   10.81

-->
```

RESULT:

Impedance matching using single stub matching is optimized by calculating the location and length of the stub for varying values of characteristic impedance keeping the load impedance constant .